

25 October 2025

CODE

-- 1. INNER JOIN: Orders with Customer and Product Names

-- Expected: OrderID, OrderDate, CustomerName, ProductName, Quantity

```
SELECT
    o.OrderID,
    o.OrderDate,
    c.CustomerName,
    p.ProductName,
    o.Quantity
FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o
INNER JOIN CUSTOMERS_LARGE AS c
    ON o.CustomerID = c.CustomerID
INNER JOIN PRODUCTS_LARGE AS p
    ON o.ProductID = p.ProductID;
```

OUTPUT

	ORDERID	ORDERDATE	CUSTOMERNAME	PRODUCTNAME	Q
12	12	2023-06-19	Customer_1248	Product_2115	
13	13	2023-04-04	Customer_1315	Product_2060	
14	14	2024-11-26	Customer_1348	Product_2144	
15	15	2023-02-23	Customer_1310	Product_2105	
16	16	2024-04-15	Customer_1219	Product_2082	
17	17	2024-04-29	Customer_1199	Product_2047	
18	18	2023-05-12	Customer_1191	Product_2101	
19	19	2024-11-12	Customer_1364	Product_2074	
20	20	2023-06-08	Customer_1333	Product_2117	
21	21	2023-08-14	Customer_1269	Product_2119	
22	22	2024-06-12	Customer_1082	Product_2053	
23	23	2023-09-17	Customer_1128	Product_2075	
24	24	2023-07-20	Customer_1181	Product_2159	
25	25	2024-01-14	Customer_1260	Product_2124	

CODE

-- 2. INNER JOIN: Customers Who Placed Orders

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate

FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c

INNER JOIN ORDERS_LARGE AS o

ON c.CustomerID = o.CustomerID;

OUTPUT

	...	CUSTOMERID	CUSTOMERNAME	COUNTRY	ORDERID	ORDERDATE
12		1248	Customer_1248	Germany	12	2023-06-19
13		1315	Customer_1315	Canada	13	2023-04-04
14		1348	Customer_1348	USA	14	2024-11-26
15		1310	Customer_1310	Germany	15	2023-02-23
16		1219	Customer_1219	Germany	16	2024-04-15
17		1199	Customer_1199	USA	17	2024-04-29
18		1191	Customer_1191	Germany	18	2023-05-12
19		1364	Customer_1364	Germany	19	2024-11-12
20		1333	Customer_1333	USA	20	2023-06-08
21		1269	Customer_1269	UK	21	2023-08-14
22		1082	Customer_1082	Germany	22	2024-06-12
23		1128	Customer_1128	UK	23	2023-09-17
24		1181	Customer_1181	India	24	2023-07-20
25		1260	Customer_1260	UK	25	2024-01-14

CODE

-- 3. LEFT JOIN: All Customers and Their Orders and include customers who have NOT placed any orders.

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID, Quantity

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate,

o.ProductID,

o.Quantity

FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c

LEFT JOIN ORDERS_LARGE AS o

ON c.CustomerID = o.CustomerID;

OUTPUT

	CUSTOMERID	CUSTOMERNAME	...	COUNTRY	ORDERID	ORDERDATE	PRODUCTID	QUANTITY
12	1248	Customer_1248		Germany	12	2023-06-19	2115	6
13	1315	Customer_1315		Canada	13	2023-04-04	2060	10
14	1348	Customer_1348		USA	14	2024-11-26	2144	7
15	1310	Customer_1310		Germany	15	2023-02-23	2105	5
16	1219	Customer_1219		Germany	16	2024-04-15	2082	3
17	1199	Customer_1199		USA	17	2024-04-29	2047	5
18	1191	Customer_1191		Germany	18	2023-05-12	2101	7
19	1364	Customer_1364		Germany	19	2024-11-12	2074	7
20	1333	Customer_1333		USA	20	2023-06-08	2117	1
21	1269	Customer_1269		UK	21	2023-08-14	2119	8
22	1082	Customer_1082		Germany	22	2024-06-12	2053	9
23	1128	Customer_1128		UK	23	2023-09-17	2075	5
24	1181	Customer_1181		India	24	2023-07-20	2159	3
25	1260	Customer_1260		UK	25	2024-01-14	2124	6

CODE

-- 4. LEFT JOIN: Product Order Count

-- Expected: ProductID, ProductName, TotalOrders

SELECT

p.ProductID,

p.ProductName,

COUNT(o.OrderID) AS TotalOrders

FROM PRACTICAL2.JOINS.PRODUCTS_LARGE AS p

LEFT JOIN ORDERS_LARGE AS o

ON p.ProductID = o.ProductID

GROUP BY p.ProductID, p.ProductName

ORDER BY TotalOrders DESC;

OUTPUT

	PRODUCTID	PRODUCTNAME	TOTALORDERS
1	2076	Product_2076	34
2	2041	Product_2041	34
3	2011	Product_2011	32
4	2135	Product_2135	32
5	2098	Product_2098	30
6	2109	Product_2109	29
7	2074	Product_2074	28
8	2197	Product_2197	28
9	2167	Product_2167	27
10	2160	Product_2160	27
11	2088	Product_2088	27
12	2147	Product_2147	27
13	2020	Product_2020	27
14	2136	Product_2136	27

CODE

```
-- 5. RIGHT JOIN: Orders with Product Info (Include Products Not Ordered)
-- Expected: OrderID, OrderDate, ProductID, ProductName, Price, Quantity
-- NOTE: Snowflake supports RIGHT JOIN natively.
```

```
-----

SELECT
    o.OrderID,
    o.OrderDate,
    p.ProductID,
    p.ProductName,
    p.Price,
    o.Quantity
FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o
RIGHT JOIN PRODUCTS_LARGE AS p
    ON o.ProductID = p.ProductID;
```

OUTPUT

	ORDERID	ORDERDATE	PRODUCTID	PRODUCTNAME	...	PRICE	QUANTITY
1	1	2023-06-10	2014	Product_2014		522	10
2	2	2023-12-07	2004	Product_2004		1996	5
3	3	2024-10-26	2171	Product_2171		76	9
4	4	2023-02-17	2007	Product_2007		156	2
5	5	2024-11-06	2061	Product_2061		1595	2
6	6	2024-11-23	2190	Product_2190		1755	3
7	7	2023-07-29	2099	Product_2099		1674	8
8	8	2023-12-06	2078	Product_2078		333	7
9	9	2025-01-25	2043	Product_2043		1947	7
10	10	2023-07-19	2141	Product_2141		1599	3
11	11	2024-11-21	2090	Product_2090		1662	1
12	12	2023-06-19	2115	Product_2115		1585	6
13	13	2023-04-04	2060	Product_2060		103	10
14	14	2024-11-26	2144	Product_2144		1284	7

CODE

-- 6. RIGHT JOIN: Customer Info with Orders (Include All Customers)

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID, Quantity

SELECT

c.CustomerID,
 c.CustomerName,
 c.Country,
 o.OrderID,
 o.OrderDate,
 o.ProductID,
 o.Quantity

FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o

RIGHT JOIN CUSTOMERS_LARGE AS c

ON o.CustomerID = c.CustomerID;

OUTPUT

	CUSTOMERID	CUSTOMERNAME	COUNTRY ...	ORDERID	ORDERDATE	PRODUCTID	QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2
6	1185	Customer_1185	Australia	6	2024-11-23	2190	3
7	1011	Customer_1011	Germany	7	2023-07-29	2099	8
8	1322	Customer_1322	Australia	8	2023-12-06	2078	7
9	1224	Customer_1224	Australia	9	2025-01-25	2043	7
10	1010	Customer_1010	UK	10	2023-07-19	2141	3
11	1324	Customer_1324	India	11	2024-11-21	2090	1
12	1248	Customer_1248	Germany	12	2023-06-19	2115	6
13	1315	Customer_1315	Canada	13	2023-04-04	2060	10
14	1348	Customer_1348	USA	14	2024-11-26	2144	7

CODE

-- 7. FULL OUTER JOIN: All Customers and All Orders

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID, Quantity

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate,

o.ProductID,

o.Quantity

FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c

FULL OUTER JOIN ORDERS_LARGE AS o

ON c.CustomerID = o.CustomerID;

OUTPUT

	CUSTOMERID	CUSTOMERNAME	COUNTRY	ORDERID	ORDERDATE	... PRODUCTID	QUANTITY
1	1251	Customer_1251	Germany	1	2023-06-10	2014	10
2	1236	Customer_1236	Australia	2	2023-12-07	2004	5
3	1170	Customer_1170	Germany	3	2024-10-26	2171	9
4	1344	Customer_1344	Canada	4	2023-02-17	2007	2
5	1319	Customer_1319	USA	5	2024-11-06	2061	2
6	1185	Customer_1185	Australia	6	2024-11-23	2190	3
7	1011	Customer_1011	Germany	7	2023-07-29	2099	8
8	1322	Customer_1322	Australia	8	2023-12-06	2078	7
9	1224	Customer_1224	Australia	9	2025-01-25	2043	7
10	1010	Customer_1010	UK	10	2023-07-19	2141	3
11	1324	Customer_1324	India	11	2024-11-21	2090	1
12	1248	Customer_1248	Germany	12	2023-06-19	2115	6
13	1315	Customer_1315	Canada	13	2023-04-04	2060	10
14	1348	Customer_1348	USA	14	2024-11-26	2144	7

CODE

-- 8. FULL OUTER JOIN: All Products and Orders

-- Expected: ProductID, ProductName, Price, OrderID, OrderDate, CustomerID, Quantity

SELECT

p.ProductID,

p.ProductName,

p.Price,

o.OrderID,

o.OrderDate,

o.CustomerID,

o.Quantity

FROM PRACTICAL2.JOINS.PRODUCTS_LARGE AS p

FULL OUTER JOIN ORDERS_LARGE AS o

ON p.ProductID = o.ProductID;

OUTPUT

	PRODUCTID	PRODUCTNAME	PRICE	ORDERID	ORDERDATE	... CUSTOMERID	QUANTITY
1	2014	Product_2014	522	1	2023-06-10	1251	10
2	2004	Product_2004	1996	2	2023-12-07	1236	5
3	2171	Product_2171	76	3	2024-10-26	1170	9
4	2007	Product_2007	156	4	2023-02-17	1344	2
5	2061	Product_2061	1595	5	2024-11-06	1319	2
6	2190	Product_2190	1755	6	2024-11-23	1185	3
7	2099	Product_2099	1674	7	2023-07-29	1011	8
8	2078	Product_2078	333	8	2023-12-06	1322	7
9	2043	Product_2043	1947	9	2025-01-25	1224	7
10	2141	Product_2141	1599	10	2023-07-19	1010	3
11	2090	Product_2090	1662	11	2024-11-21	1324	1
12	2115	Product_2115	1585	12	2023-06-19	1248	6
13	2060	Product_2060	103	13	2023-04-04	1315	10
14	2144	Product_2144	1284	14	2024-11-26	1348	7

ALL CODE

-- 1. INNER JOIN: Orders with Customer and Product Names

-- Expected: OrderID, OrderDate, CustomerName, ProductName, Quantity

SELECT

o.OrderID,

o.OrderDate,

c.CustomerName,

p.ProductName,

o.Quantity

FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o

INNER JOIN CUSTOMERS_LARGE AS c

ON o.CustomerID = c.CustomerID

INNER JOIN PRODUCTS_LARGE AS p

ON o.ProductID = p.ProductID;

-- 2. INNER JOIN: Customers Who Placed Orders

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate

FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c

INNER JOIN ORDERS_LARGE AS o

ON c.CustomerID = o.CustomerID;

-- 3. LEFT JOIN: All Customers and Their Orders and include customers who have NOT placed any orders.

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID, Quantity

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate,

o.ProductID,

o.Quantity

```
FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c
LEFT JOIN ORDERS_LARGE AS o
  ON c.CustomerID = o.CustomerID;
```

```
-----
-- 4. LEFT JOIN: Product Order Count
-- Expected: ProductID, ProductName, TotalOrders
-----
```

```
SELECT
  p.ProductID,
  p.ProductName,
  COUNT(o.OrderID) AS TotalOrders
FROM PRACTICAL2.JOINS.PRODUCTS_LARGE AS p
LEFT JOIN ORDERS_LARGE AS o
  ON p.ProductID = o.ProductID
GROUP BY p.ProductID, p.ProductName
ORDER BY TotalOrders DESC;
```

```
-----
-- 5. RIGHT JOIN: Orders with Product Info (Include Products Not Ordered)
-- Expected: OrderID, OrderDate, ProductID, ProductName, Price, Quantity
-----
```

```
SELECT
  o.OrderID,
  o.OrderDate,
  p.ProductID,
  p.ProductName,
```

p.Price,

o.Quantity

FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o

RIGHT JOIN PRODUCTS_LARGE AS p

ON o.ProductID = p.ProductID;

-- 6. RIGHT JOIN: Customer Info with Orders (Include All Customers)

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID,
Quantity

SELECT

c.CustomerID,

c.CustomerName,

c.Country,

o.OrderID,

o.OrderDate,

o.ProductID,

o.Quantity

FROM PRACTICAL2.JOINS.ORDERS_LARGE AS o

RIGHT JOIN CUSTOMERS_LARGE AS c

ON o.CustomerID = c.CustomerID;

-- 7. FULL OUTER JOIN: All Customers and All Orders

-- Expected: CustomerID, CustomerName, Country, OrderID, OrderDate, ProductID,
Quantity

```
-----  
SELECT  
    c.CustomerID,  
    c.CustomerName,  
    c.Country,  
    o.OrderID,  
    o.OrderDate,  
    o.ProductID,  
    o.Quantity  
FROM PRACTICAL2.JOINS.CUSTOMERS_LARGE AS c  
FULL OUTER JOIN ORDERS_LARGE AS o  
    ON c.CustomerID = o.CustomerID;
```

```
-----  
-- 8. FULL OUTER JOIN: All Products and Orders  
-- Expected: ProductID, ProductName, Price, OrderID, OrderDate, CustomerID, Quantity
```

```
-----  
SELECT  
    p.ProductID,  
    p.ProductName,  
    p.Price,  
    o.OrderID,  
    o.OrderDate,  
    o.CustomerID,  
    o.Quantity  
FROM PRACTICAL2.JOINS.PRODUCTS_LARGE AS p  
FULL OUTER JOIN ORDERS_LARGE AS o
```

ON p.ProductID = o.ProductID;